Context Promotion

Uncertainty-Aware Hybrid Retrieval for Long-Document RAG
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What the paper proposes

The paper argues that fine chunks can locate evidence while broader neighboring context improves coherence. Its parent-promotion variant uses fine-grained hits to return broader local context.

What changed in this project

Added a same-document adjacent-page support score. A chunk receives a small bonus when nearby pages in the same manual are semantically compatible, promoting locally coherent evidence without changing the source text.

Measured results

Metric	Baseline	Method	Change
Hit@1	80.0%	80.0%	+0.0 pp
Hit@5	85.0%	90.0%	+5.0 pp
Recall@5	64.7%	67.2%	+2.5 pp
NDCG@10	84.0%	84.4%	+0.4 pp
MRR@20	82.9%	83.2%	+0.4 pp
Latency	26.9 ms	3.6 ms	—

Inference

This was one of the practical wins: Hit@5 rose from 85.0% to 90.0%, Recall@5 from 64.7% to 67.2%, NDCG@10 from 84.0% to 84.4%, and MRR@20 from 82.9% to 83.3%. Mean latency fell to 3.6 ms for the reranking step.